# GS Meter

User Manual

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## GS Meter Manual

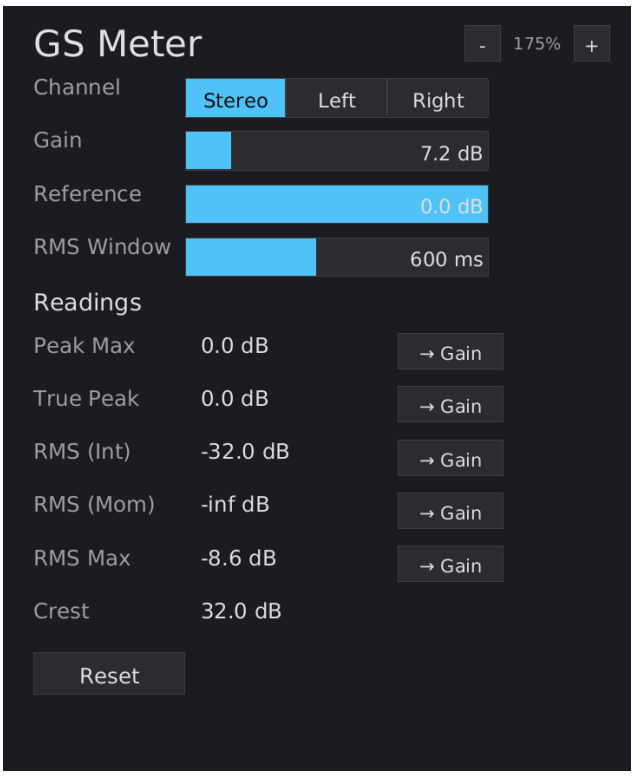


Figure 1: GS Meter interface

### What is GS Meter?

GS Meter is a lightweight loudness meter with an integrated gain utility, purpose-built for clip-to-zero workflows. It is designed to replace dpMeter5 in projects where you need a meter on every track, bus, and master – 50+ instances without killing your CPU or eating your RAM.

**If you're not doing clip-to-zero, this plugin is probably not for you.** It has no visual meters, no bar graphs, no loudness history. It shows numbers and lets you set gain from them. That's it.

### Why does this exist?

dpMeter5 is the standard meter for clip-to-zero, but running 50+ instances of any full-featured meter adds up. GS Meter is built from the ground up for density:

- **CPU rendering** – no OpenGL, no GPU driver loaded. Zero memory overhead when the GUI is closed.
- **SIMD metering** – peak detection and RMS accumulation use f32x16 SIMD, true peak FIR uses SIMD dot products.
- **50 instances: ~16% CPU, 48 MB total memory** (measured at 48 kHz / 1024 samples, GUI closed).

## Installation

Build from source (requires nightly Rust):

```
cargo nih-plug bundle gs-meter --release
```

The bundler outputs to `target/bundled/`. Copy either the `.vst3` or `.clap` file (you only need one – use whichever your DAW supports) to your plugin directory:

- **Linux**: `~/.vst3/` or `~/.clap/`
- **macOS**: `~/Library/Audio/Plug-Ins/VST3/` or `~/Library/Audio/Plug-Ins/CLAP/`
- **Windows**: `C:\Program Files\Common Files\VST3\` or `C:\Program Files\Common Files\CLAP\`

## Controls

### Gain

Output gain applied to the audio signal. Range: -40 to +40 dB. Default: 0 dB.

This is the gain you adjust to match your reference level. The "-> Gain" buttons next to each reading auto-set this value.

### Reference

Your target level. Range: -60 to 0 dB. Default: 0 dB.

When you click a "-> Gain" button, the gain is set to: `reference - meter_reading`. For example, if your reference is -14 dB and the integrated RMS reads -18 dB, clicking "-> Gain" sets gain to +4 dB.

### RMS Window

The sliding window size for momentary RMS measurement. Range: 50 to 3000 ms. Default: 600 ms.

This matches dpMeter5's RMS Window setting. Changing it resets the momentary max.

### Channel Mode

- **Stereo** – readings sum both channels' power (matches dpMeter5 SUM mode)
- **Left** – readings from the left channel only
- **Right** – readings from the right channel only

### Scaling

Use the **-** / **+** buttons in the upper right corner, or **Ctrl+=** / **Ctrl+-** on the keyboard. Range: 75% to 300%.

## Readings

### Peak Max

Highest absolute sample value since reset, in dBFS.

### True Peak

Highest inter-sample peak since reset, in dBTP. Uses 4x polyphase oversampling with the exact ITU-R BS.1770-4 Annex 2 reference filter coefficients (48-tap, 4-phase FIR). At sample rates >=96 kHz, uses 2x oversampling. At >=192 kHz, true peak equals sample peak (no interpolation needed).

True peak may read slightly different from dpMeter5 (typically within 0.1-0.2 dB). This is due to filter design differences – both implementations comply with the ITU standard.

### RMS (Int)

Integrated RMS since reset. Accumulates the mean-square of all samples processed and reports the square root. Uses f64 accumulation for precision over long measurement periods.

In Stereo mode, this is `sqrt(ms_L + ms_R)` – the sum of both channels' mean-square power, matching dpMeter5's SUM mode.

### RMS (Mom)

Momentary RMS over the current sliding window (set by RMS Window). Updates continuously.

### RMS Max

Highest momentary RMS value observed since reset. Updated once per audio buffer.

### Crest

Crest factor: `Peak Max (dB) - RMS Integrated (dB)`. Shows the dynamic range of the signal – higher values mean more transient peaks relative to average loudness. This is a live value that settles as the integrated RMS accumulates.

In Stereo mode, crest factor uses the stereo peak (max of L/R) and stereo RMS (sum of power). This matches dpMeter5's convention but gives values ~3 dB lower than a per-channel crest factor for balanced stereo content.

### -> Gain Buttons

Each reading (except Crest) has a "-> Gain" button that sets: `Gain = Reference - Reading`.

Use this after letting the meter accumulate for a representative section of audio. The integrated RMS button is the most useful for clip-to-zero – it tells you how much gain to add to hit your reference level.

### Reset

Clears all accumulated values: peak max, true peak, integrated RMS, momentary max, and crest factor. Momentary RMS continues from the current window contents.

**Double-click** any slider or the channel selector to reset it to its default value.

## Clip-to-Zero Workflow

1. Insert GS Meter on every track and bus (it's cheap – 50 instances use 16% CPU)
2. Set your reference level to 0 dB (the default)
3. Play a representative section of the song
4. Click the "-> Gain" button next to Peak Max to auto-set the gain
5. Reset and repeat if needed

## Technical Notes

- **No audio-thread allocations** – the process() callback never allocates heap memory
- **Pre-allocated RMS ring buffer** – sized for 3000 ms at 192 kHz (worst case), no resize on the audio thread
- **CPU rendering** – uses tiny-skia (software rasterizer) + fontdue (glyph cache) + softbuffer (pixel buffer). No OpenGL context, no GPU drivers loaded
- **SIMD** – uses Rust's portable SIMD (`std::simd::f32x16`) for peak detection, sum-of-squares, and true peak FIR convolution
- **Embedded font** – DejaVu Sans, compiled into the binary. No runtime font loading

## Formats

- CLAP
- VST3
- Standalone (JACK or ALSA backend)

## License

GPL-3.0-or-later